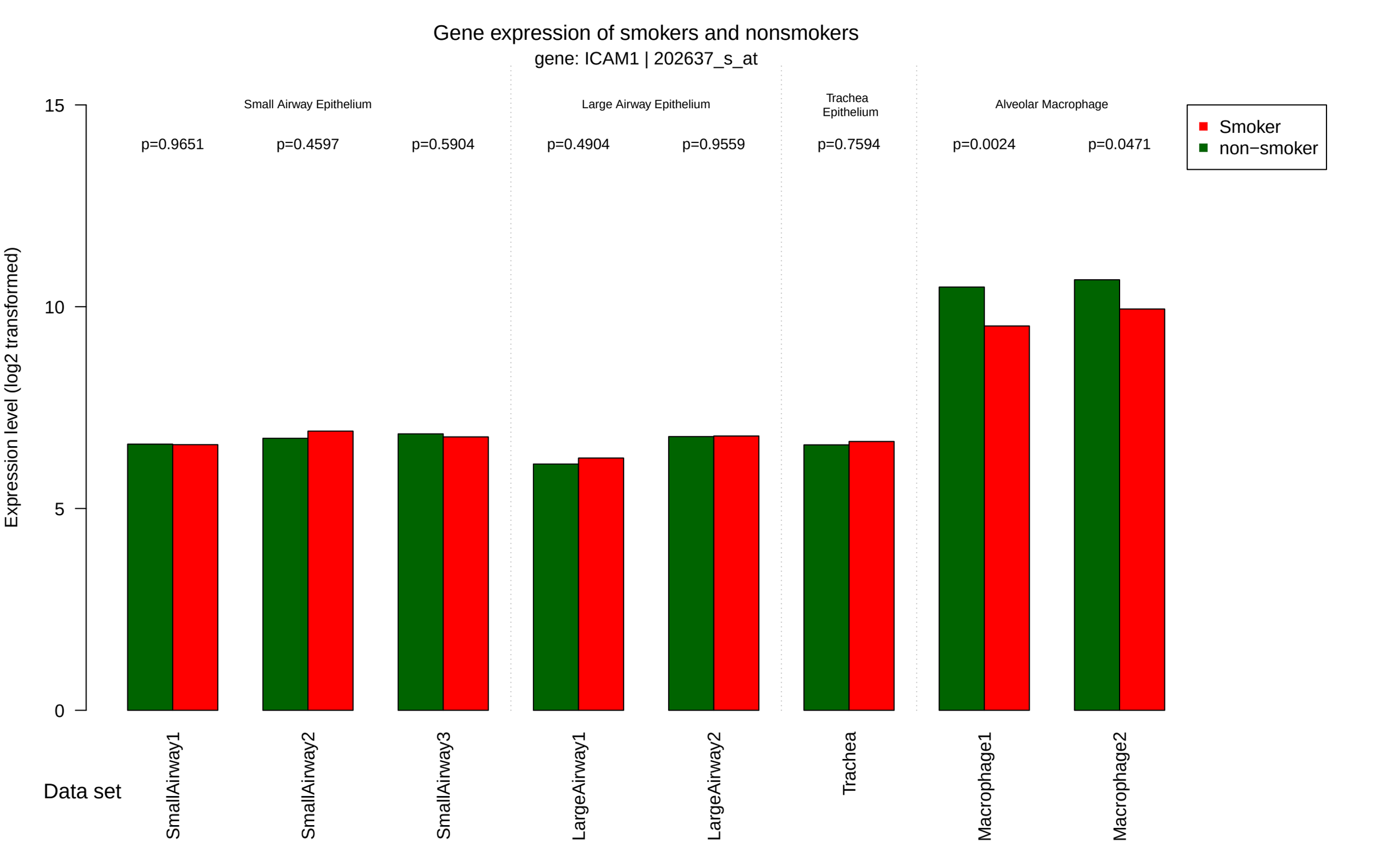
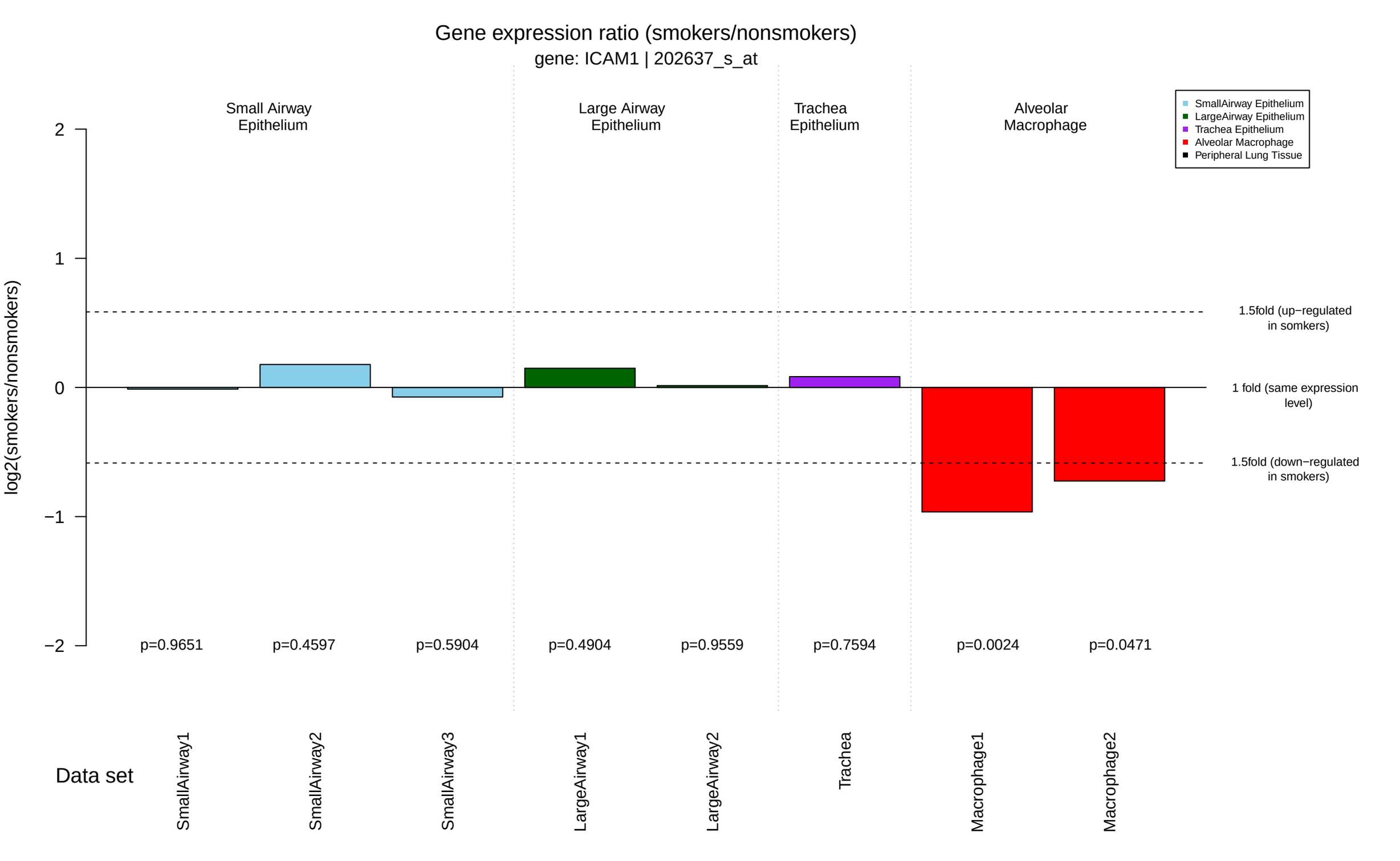
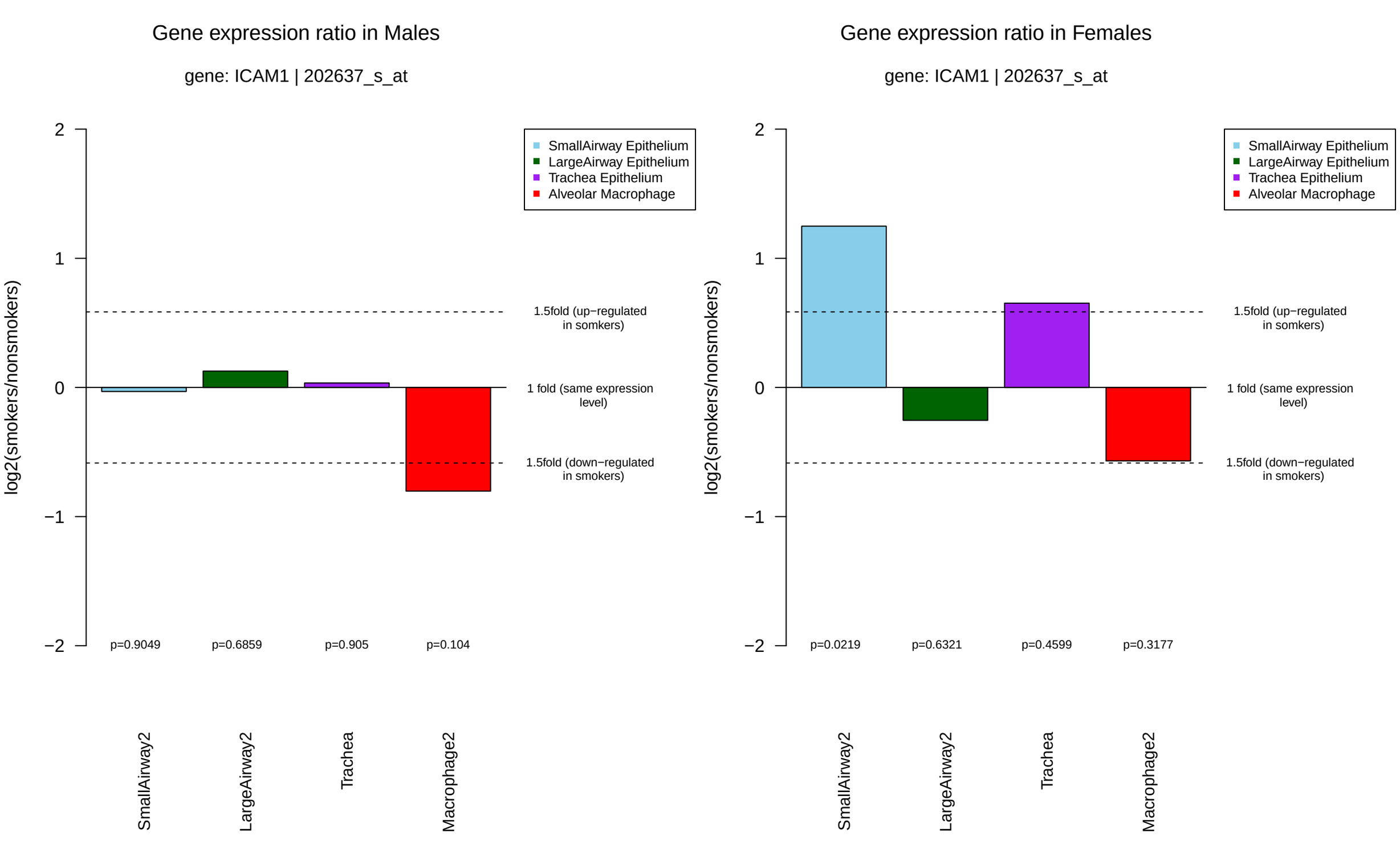
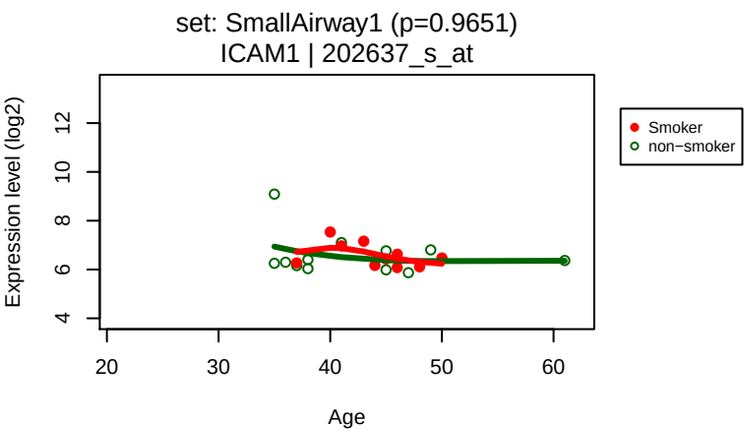
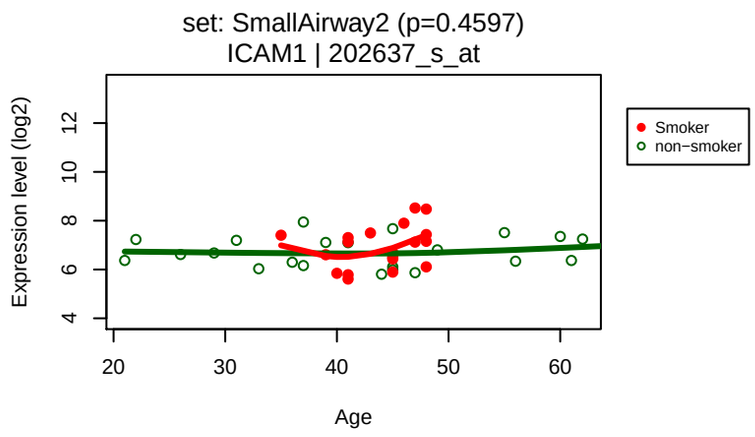
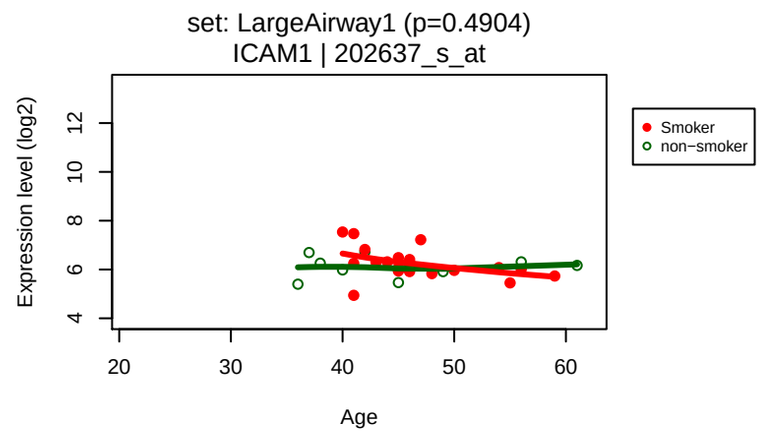

Gene expression of smokers and nonsmokers
gene: ICAM1 | 202637_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9651
p=0.4597
p=0.5904
p=0.4904
p=0.9559
p=0.7594
p=0.0024
p=0.0471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

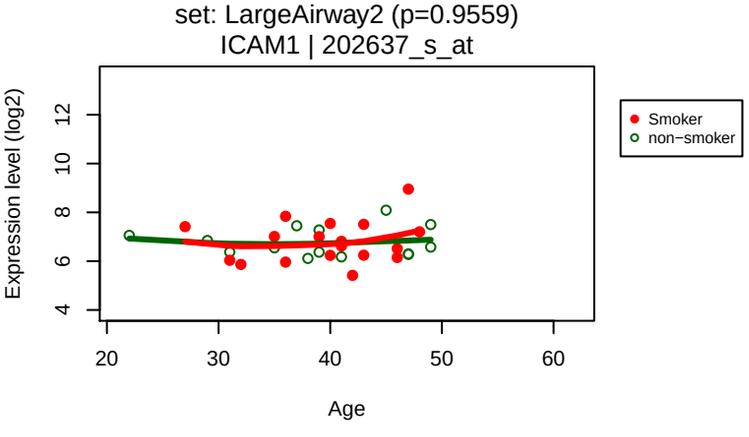

Gene expression ratio (smokers/nonsmokers)
gene: ICAM1 | 202637_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9651
p=0.4597
p=0.5904
p=0.4904
p=0.9559
p=0.7594
p=0.0024
p=0.0471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

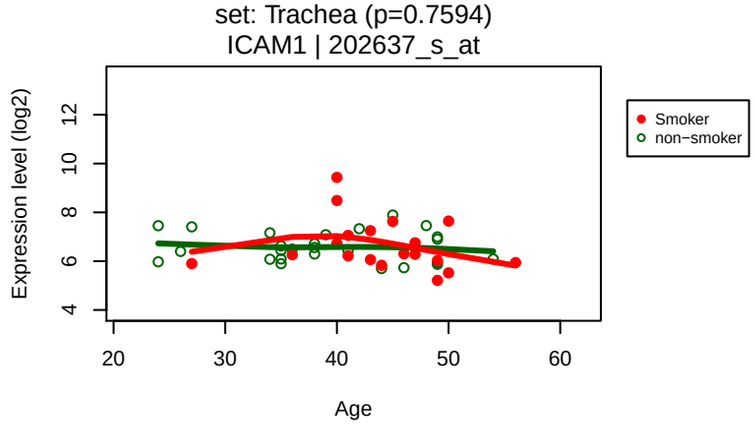

Gene expression ratio in Males
gene: ICAM1 | 202637_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9049
p=0.6859
p=0.905
p=0.104
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICAM1 | 202637_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0219
p=0.6321
p=0.4599
p=0.3177
SmallAirway2
LargeAirway2
Trachea
Macrophage2

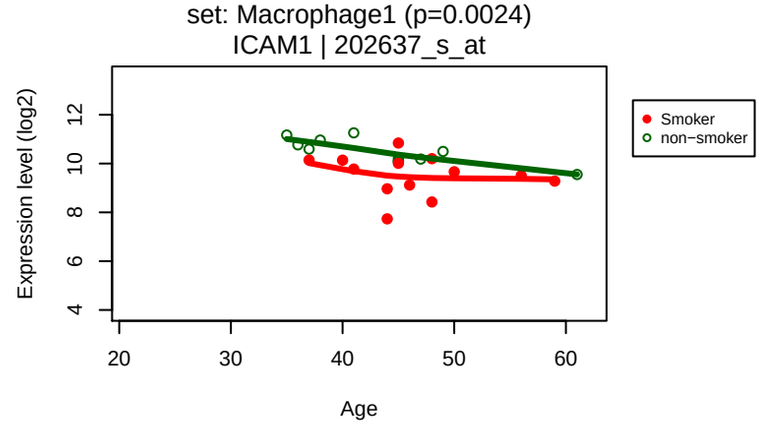

set: SmallAirway1 (p=0.9651)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker

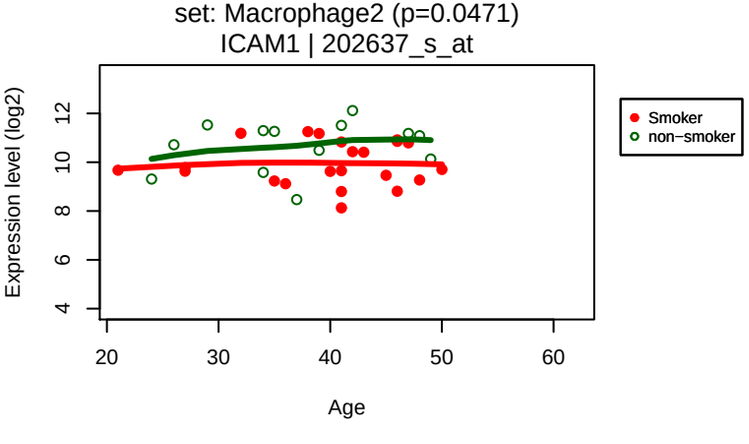

set: SmallAirway2 (p=0.4597)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker

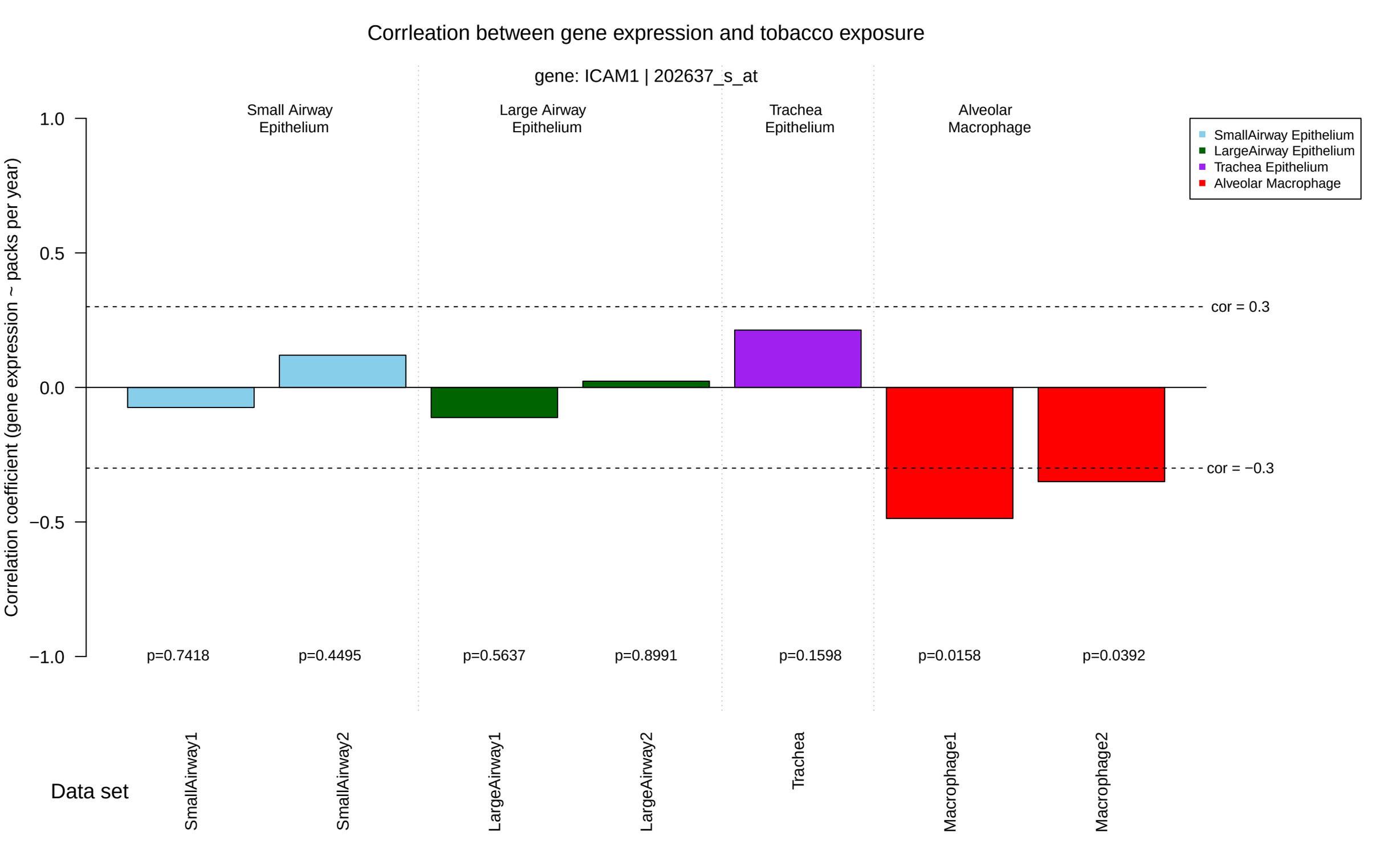

set: LargeAirway1 (p=0.4904)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9559)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7594)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0024)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0471)
ICAM1 | 202637_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICAM1 | 202637_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7418
p=0.4495
p=0.5637
p=0.8991
p=0.1598
p=0.0158
p=0.0392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2